THE UNIVERSITY OF HONG KONG
DEPARTMENT OF MATHEMATICS

MATH2241 Introduction to Mathematical Analysis
Sample Test 1 Solutions

1. We shall prove that $\lim_{n\to\infty} \frac{n-1}{n^2-3} = 0$. For any $\varepsilon > 0$, we choose $N = \left\lceil \frac{1}{\varepsilon} + 3 \right\rceil$. Then for any $n \geqslant N > 3$, we have

$$\left| \frac{n-1}{n^2-3} - 0 \right| = \frac{n-1}{n^2-3} < \frac{n}{n^2-3n} = \frac{1}{n-3} \leqslant \frac{1}{N-3} \leqslant \varepsilon.$$

This implies $\lim_{n\to\infty} \frac{n-1}{n^2-3} = 0$ by definition.

2. (a) For any $x \in X$, there exists $y \in Y$ such that $x \leqslant y$. Also, we have $y \leqslant \sup Y$. This yields $x \leqslant \sup Y$. Since this holds for any $x \in X$, $\sup Y$ is an upper bound for X. Therefore, $\sup X \leqslant \sup Y$ as $\sup X$ is the smallest upper bound.

   (b) No. Consider $X = \{1, 3\}$ and $Y = \{0, 4\}$. Then for each $x \in X$, we can pick $y = 4 \in Y$ such that $x < y$. However, $\inf X = 1$ is larger than $\inf Y = 0$.

3. (a) Since $\{b_1, b_2, \ldots, b_k\}$ is a finite set, its minimum exists. The infimum c_k is just this minimum value.

   (b) As $\{b_1, b_2, \ldots, b_k\}$ is a subset of $\{b_1, b_2, \ldots, b_{k+1}\}$, we have $c_k \geqslant c_{k+1}$ for any $k \in \mathbb{N}$. This shows $\{c_k\}$ is a monotonic decreasing sequence. Given that $\{b_n\}$ is bounded, $\{c_k\}$ is also bounded as each c_k is just some of the b_i's. Therefore, $\{c_k\}$ is convergent by the monotone convergence theorem.

   (c) No. Consider $\{b_n\}$ where $b_1 = 0$ and $b_n = 1$ for any $n \geqslant 2$. Then $c_k = 0$ for each $k \in \mathbb{N}$ and hence $\lim_{k\to\infty} c_k = 0$. However, we have $\liminf_{n\to\infty} b_n = \lim_{n\to\infty} \{0, 1, 1, \ldots\} = 1 \neq 0$.

4. Since a_1 and b_1 are positive, it is obvious that a_n and b_n are positive by induction. This implies $a_{n+1} = a_n + b_n > a_n$. So $\{a_n\}$ is an increasing sequence. By the monotone convergence theorem, $\{a_n\}$ is convergent if and only if it is bounded above. We label the statements (i), (ii) and (iii) in that order.

   - (iii) $\Rightarrow$ (i): Obvious from the above discussion.
   - (i) $\Rightarrow$ (ii): If $\{a_n\}$ converges to L, then from $b_n = a_{n+1} - a_n$, we know that $\{b_n\}$ converges to $L - L = 0$.
   - (ii) $\Rightarrow$ (iii): Suppose on the contrary that $\{b_n\}$ is convergent but $a_N > 1$ for some $N \in \mathbb{N}$. This implies $a_n \geqslant a_N > 1$ and $b_{n+1} = a_n b_n > b_n$ for any $n \geqslant N$. Therefore $\{b_n\}$ is eventually monotonic increasing. It follows that the limit L' of $\{b_n\}$ must be at least b_N.

     Since each b_n is nonzero and the limit L' is nonzero, from $a_n = \frac{b_{n+1}}{b_n}$, we know that $\{a_n\}$ converges to $\frac{L'}{L'} = 1$. But we have shown that $\{a_n\}$ is increasing with $a_N > 1$, so its limit is at least a_N. This is a contradiction.

5. (a) False. Consider $S = \left\{\frac{1}{m} : m \in \mathbb{N}\right\}$. Since $0 \leqslant \frac{1}{m} \leqslant 1$ for any $m \in \mathbb{N}$, S is bounded. Clearly, it is an infinite set. Also, $\sup S = 1$ as 1 is an upper bound of S and it belongs to S. If we take $n = \frac{1}{3}$, there is no element a in S satisfying $1 - \frac{1}{3} < a < 1$.

   (b) False. Consider $\{a_n\}$ and $\{b_n\}$ defined by $a_n = (-1)^{n+1}$ and $b_n = (-1)^n$ for any $n \in \mathbb{N}$. Then both $\{a_n\}$ and $\{b_n\}$ have constant subsequences with limit 1, but $\{a_n + b_n\}$ is the zero sequence, which has no subsequence with limit $1 + 1 = 2$.

   (c) True. Suppose on the contrary that $\{c_n\}$ converges to $L \in \mathbb{R}$. Then the sequence is bounded. Suppose $c_n \leqslant M$ for any $n \in \mathbb{N}$. Then the interval $(M, M + 1)$ contains no element in C. This contradicts the fact that C is dense.

   Remark. Such a divergent sequence $\{c_n\}$ exists. For example, recall that $\mathbb{Q}$ is a countable set. Therefore, we can label all elements in $\mathbb{Q}$ as $c_1, c_2, \ldots$. This gives an example for the sequence $\{c_n\}$ such that $C = \mathbb{Q}$ is dense.